AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 25 · TEACHER GUIDE

Generate, Retrieve, or Calculate?

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Teach the operation and the check together.

Model the current notice, complete expressions and claim-by-claim review, then reserve record B for independent evidence.

Keep one habit: Look for inspectable sources, correct operations and honest unknowns.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Distinguish retrieving a record, calculating from **input**s and generating wording

Keep operations distinct without presenting three mutually exclusive product categories. A modern system can **retrieve** records, call a calculator and **generate** a final response.

Lewis and colleagues provide a primary example combining **retrieval** and **generation**: **Retrieval**-Augmented **Generation** for Knowledge-Intensive NLP Tasks (2020), https://arxiv.org/abs/2005.11401.

Use this as teacher background, not required pupil reading. The fictional notices and arithmetic here are independent classroom examples.

By the end, I can separate **retrieval**, **calculation** and **generate**d wording.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- The fictional N1 and N2 notice details printed in the student lesson
- Paper for three-column **trace**s labelled source facts, **calculation** and wording
- The independent library record B in W4; no online account is needed

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Privacy by Choosing Less. Give an example and a mistake that the idea helps you avoid.

Look for: Justify which information a stated task needs and remove unnecessary fields

Reference example, not the only acceptable wording: Keep topic choice; remove name, full date of birth, address and photograph. The totals do not require identity or contact details.

Collecting extra fields creates copies and exposure to manage even if those fields are later ignored or deleted.

Without opening the old lesson, explain one key idea from Missing and Unbalanced Examples. Give an example and a mistake that the idea helps you avoid.

Look for: Name the target collection, selection list and cases actually assessed

Reference example, not the only acceptable wording: The target is all 24 specified cards across four conditions. The sample is the 6 bright-circle cards. Dim circles, bright squares and dim squares are absent.

A frame containing only bright-circle IDs excludes every other condition, regardless of how randomly IDs are drawn from it.

Without opening the old lesson, explain one key idea from Similarity. Give an example and a mistake that the idea helps you avoid.

Look for: **Calculate** similarity using explicitly selected features and a stated scoring rule

Reference example, not the only acceptable wording: A: 1, 1, 0 gives 2. B: 0, 0, 1 gives 1. C: 1, 0, 0 gives 1. A is the only highest scoring candidate among A, B and C under equal weights.

The task needs the three totals, not pupil identities. Use the least information needed and follow the teacher's approved tool and data rules.

▼ **Earlier wording, alternatives and picture transcripts**

Teach the learning cycle

Retrieval answer guidance

Privacy by Choosing Less

Missing and Unbalanced Examples

Similarity

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Three tasks are: invent a school motto, find the exact fire-drill meeting point in a supplied handbook, and divide 84 stickers among 7 teams. Choose **generation**, **retrieval** or **calculation** for each. Which **output**s need checking?

Concept to explain

Different systems create, look up, or compute answers in different ways.

Term	Meaning
generate	create a new output from learned patterns
retrieve	find and return stored information
calculate	apply defined operations to values
provenance	where an output or claim came from

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach mixed systems one inspectable operation at a time

> Model source selection, complete expressions and factual review of generated wording.

Define the operations and current notice

Three tasks can produce similar-looking text while doing different work. **Retrieval** finds existing information. **Calculation** applies mathematical operations to **input** values.

Generation produces content using a model and the context it receives. A single application can combine all three, so the label AI does not tell you which step happened.

Our fictional makers club has a current notice N2: Wednesday, Room 4, six tables with four places at each table, and three places reserved for helpers. An older notice N1 says Room 2. N2 explicitly replaces N1 for this event.

These records are invented for this lesson; no search, account or real booking is needed.

To answer “Which room does the current notice list?”, find N2 and return Room 4 with the record identifier. That is **retrieval**. To answer “How many places remain after the helper reservation?”, use 6 × 4 − 3 = 21.

That is **calculation**. To suggest a friendly invitation, **generate** wording and check that its factual details agree with N2.

Inspect source version and arithmetic meaning

Retrieving N1 accurately would still give an outdated room for this task. Successful **retrieval** means finding a record, not proving that every record is correct or current. Check the event, source and version.

Here the explicit replacement statement tells us which notice governs the classroom example.

A calculator given 6 × 4 correctly returns 24, but 24 is the total capacity, not the number remaining after three reserved places. Correct arithmetic on an incomplete expression answers the wrong question.

Explain the **input** values, units and operations before accepting a number.

Audit individual draft statements

Draft statement	Evidence in this example	Decision
Meet Wednesday in Room 4	Current notice N2	Supported by the record
There are 21 non-helper places	6 × 4 − 3 from N2	Supported by calculation
Free snacks will be provided	Not stated in N2	Remove or verify separately

A **generate**d invitation might add free snacks because that detail fits a familiar invitation pattern. Plausible wording is not **evidence** that snacks are provided.

Remove that claim or obtain an appropriate confirming source before including it. **Generate**d content can be useful and can contain correct facts; its fluent style alone does not verify those facts.

Generation also does not guarantee originality. A response may repeat wording already present in its context or elsewhere.

Retrieval and **calculation** can feed a generator, and a generator can propose a **calculation** for another tool. Keep the **evidence** path visible instead of assuming that every sentence came from the same operation.

Return to the makers-club notice N2

AI ACADEMY / AI-1/25 · 03 · INSPECT THE LESSON MODEL

One club task with three kinds of work

Operation	Example output	Required check
Retrieve N2	Room 4 on Wednesday	Right event and version
Calculate	6 × 4 − 3 = 21	Inputs and operations
Generate wording	Try a making activity	Facts and constraints

Retrieval supplies recorded information; calculation derives a result; generation drafts wording. Each operation needs a check suited to what it does.

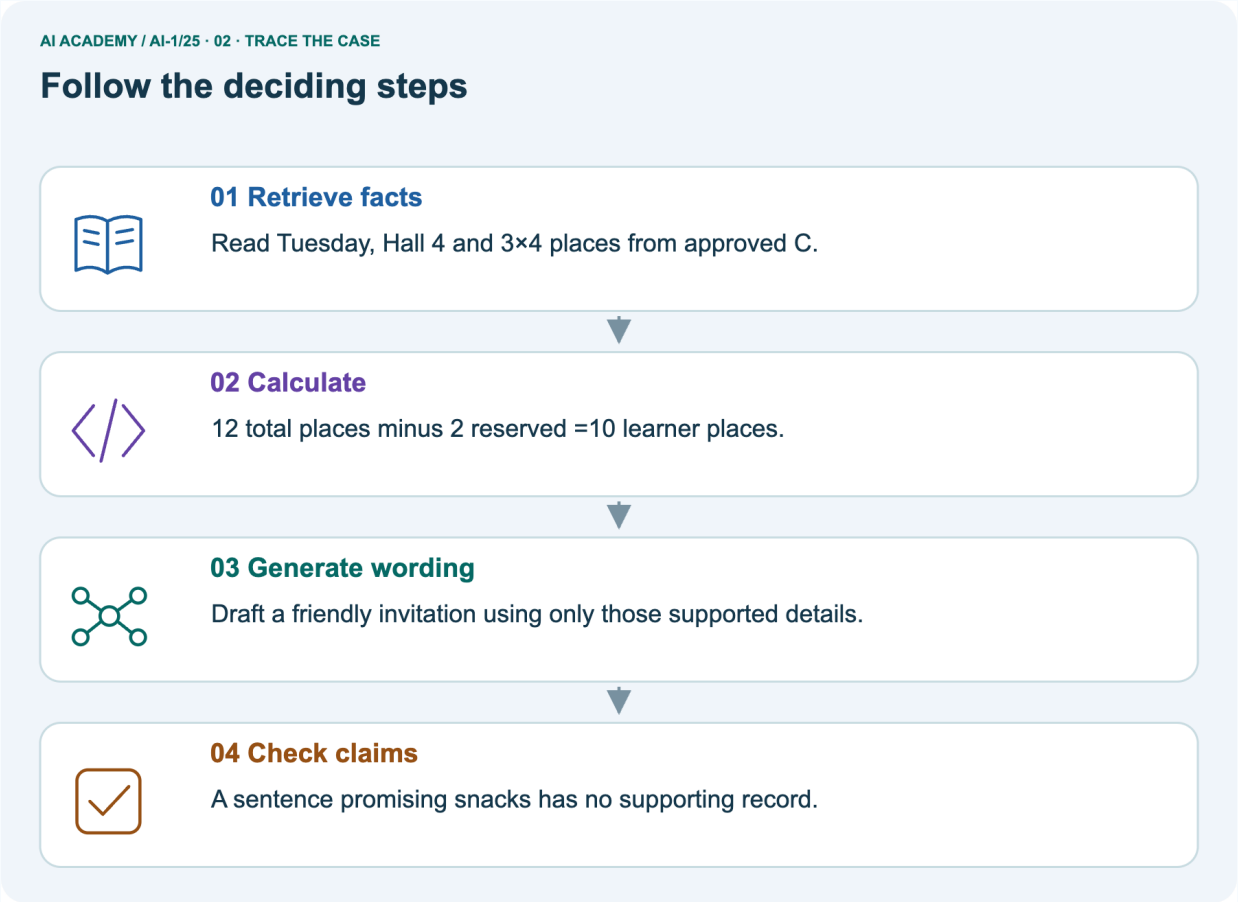
Follow the labelled picture with the matching explanation. Open the vector diagram

Trace the separate craft invitation

A fictional craft invitation combines three different operations.

Approved record C: Tuesday, Hall 4, three tables with four total places each.

Two places are reserved for helpers; no snack information is supplied.



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Retrieve facts	Read Tuesday, Hall 4 and 3×4 places from approved C.
2. Calculate	12 total places minus 2 reserved =10 learner places.
3. Generate wording	Draft a friendly invitation using only those supported details.
4. Check claims	A sentence promising snacks has no supporting record.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Retrieval copies relevant source information; **calculation** follows arithmetic; **generation** proposes wording. These jobs can cooperate, but one does not validate the others.

A friendly **generate**d sentence can still contain an invented detail. Keep the approved source beside the draft and check each factual claim separately.

Limit retrieval claims

The fictional source must itself be current and authoritative; **retrieval** alone does not ensure a record is correct.

Change only the reservation

Change: Reserve four places instead of two, keeping the room and day unchanged.

Prediction: Eight learner places remain.

Reason: 3×4−4=8; this changes capacity for learners, not the **retrieve**d date or location.

Use the picture to elicit a trace

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

<details>
<summary>▼ Earlier wording, alternatives and picture transcripts</summary>

Explain the mechanism

Follow the information instead of imagining magic inside the machine.

> **1. Input — information entering**
>
> **INPUT** question and available sources or rules

> **2. Process — the deciding step**
>
> **PROCESS generate** patterns, **retrieve** records, or **calculate**

> **3. Output — the result**
>
> **OUTPUT** answer with **mechanism**-appropriate checking

CORE IDEA Different systems create, look up, or compute answers in different ways.
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UNDER THE HOOD **Generation** synthesizes **output**s from learned distributions, **retrieval** selects stored records, and **calculation** applies formal operations.

Products often combine all three; verification should match the component that produced each claim.

Draw a fresh example with all three stages:

Every step connects a claim to observable **evidence**.

CASE A story generator creates new wording, a library search **retrieve**s a stored title, and a calculator computes a result from stated operations.

STEP 1 — OBSERVE question and available sources or rules

STEP 2 — **TRACE generate** patterns, **retrieve** records, or **calculate**

STEP 3 — CONCLUDE answer with **mechanism**-appropriate checking

MODEL REASONING Check **generate**d claims against independent **evidence**, check **retrieve**d facts against the cited record and freshness, and recompute **calculation**s or inspect the operation. Mixed **output**s need more than one check.

DO NOT OVERCLAIM Fluent wording does not reveal whether a fact was **retrieve**d, **generate**d, or **calculate**d.

What new **evidence** could change this answer?

Concrete teaching case

A learner needs a new poem, 18+7, and a passage from the class handbook.

Choose the operation for each.

Exact explanation to model

Generate the poem, **calculate** 25, **retrieve** the actual passage. Inventing a fluent passage is not **retrieval**.

Explain the added worked example

Key terms

Term	Meaning
Retrieval	Finding and returning existing information from a collection
Calculation	Applying specified mathematical operations to input values
Generation	Producing content using a model and its input context
Source record	An existing item whose content and origin can be inspected
Trace	A record of the inputs, operations and outputs used in a task

Model the reasoning aloud

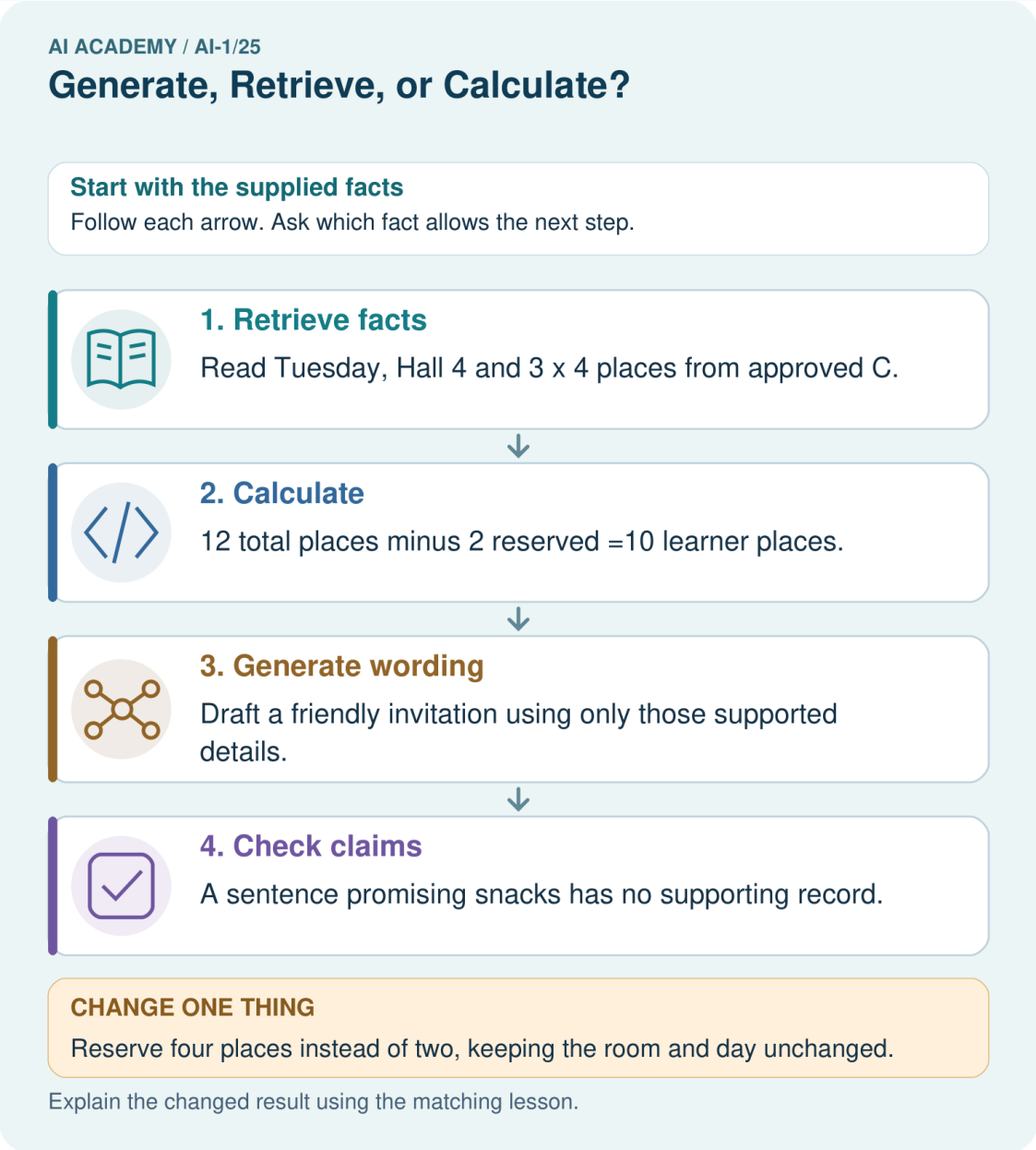
Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Generate, Retrieve, or Calculate?

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: A calculator correctly returns 24 for 6 × 4. A pupil reports this as the number of non-helper places available. Explain the problem and write the correct expression, result and meaning.

Expected reasoning and fading help

Teacher check: 24 is total capacity before the three helper places are reserved. The requested remaining count is 6 × 4 − 3 = 21 non-helper places.

The calculator can execute the supplied expression correctly while that expression omits a required step.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full **trace** only when the partial **trace** is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for **Generation retrieval** and **calculation**. Proposed explanation: “Retrieving a record proves it is current”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Check event and version; N1 is superseded by N2 in this example.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

Label each arrow with its operation. Can the system add snacks because invitations often mention them?

The arrows are **retrieval**, **calculation** and **generation**. No: snacks need confirmation; familiar wording is not **evidence**.

Changed-case reasoning: Eight learner places remain. 3×4−4=8; this changes capacity for learners, not the **retrieve**d date or location.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or **calculation** that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: A learner needs a new poem, 18+7, and a passage from the class handbook.

Write the question to explain: Choose the operation for each.

Reveal after **prediction**s: **Generate** the poem, **calculate** 25, **retrieve** the actual passage. Inventing a fluent passage is not **retrieval**.

Use the earlier changed question as optional later practice. Reserve Trace a new library workshop for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, **trace** conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Correct Thursday to Wednesday using N2; Room 4 is supported. Remove free snacks unless independently verified. Replace 24 remaining with 21 from 6 × 4 − 3.

One sentence: The makers club meets Wednesday in Room 4, with 21 places after the helper reservation. Exit: 6 × 4 − 5 = 19; day, room, table count and places per table stay the same.

The changed reservation invalidates the old remaining count.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve the minimal-input principle from Week 24.
7 to 20 minutes	Compare the three operations using N2 and complete W1.
20 to 33 minutes	Trace the diagram and complete W2. Ask what the result’s unit and meaning are.
33 to 45 minutes	Complete W3 and separate a retrieved record from a current record.
Session 2 start 0 to 8 minutes	Retrieve why correct arithmetic can still answer the wrong question.

Stage	Teacher action
8 to 23 minutes	Complete the guided correction. Require evidence for each factual claim.
23 to 38 minutes	Collect W4 independently, with source, expression, result and unsupported detail.
38 to 45 minutes	Complete W5 and the exit. Discuss identical outputs with different or unknown methods.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

For the nine paper cards, 1–3 describe retrieval: check J or K and the relevant version. Cards 4–6 describe calculation: 4×5=20 places; 20−3=17 non-helper places; 18÷3=6 cards per table. Check inputs, expression, units and arithmetic. Cards 7–9 request generated suggestions or wording; inspect factual claims, constraints and suitability. An invitation may combine retrieved facts and generated wording; classify the described main operation without claiming the real implementation from output alone. A supplied paper description is not a tool execution log. Reserve Trace a new library workshop for the individual attempt.

Run and record

1. Complete the task: Classify nine **output**s by **mechanism** and verification need.
2. Keep a normal case and an edge case clearly separated.
3. Record what happened—even when it does not match your **prediction**.
4. Explain the result, one limitation, and the next useful test.

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

Retrieving a record proves it is current: Check event and version; N1 is superseded by N2 in this example.

A correct calculator result always answers the task: The expression must include the required reservation and correct units.

Fluent **generate**d details are verified facts: Free snacks and transport need **evidence**; wording alone is insufficient.

Choose support from the response

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Predict first, test honestly, and explain what the **evidence** means.

State your **prediction** before the result is known.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Trace a new library workshop for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

CLAIM TO REPAIR Generating plausible text is the same as retrieving a fact.

For guessing, return to given facts and request a **trace**. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the **calculation**.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

The final sentence reveals the method: Matching **output**s do not establish matching operations; inspect records and **trace**s.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: Current fictional record B lists a library workshop on Friday in Room 5: five tables with six places each, with four places reserved for helpers. B replaces old record A, which lists Room 3.

A draft says “Thursday in Room 3, with 30 places remaining and a free bus home.” **Retrieve** the current day and room, **calculate** the non-helper places, write a corrected factual sentence and identify the detail that cannot be supported by B.

Retrieve Friday and Room 5 from B. **Calculate** 5 × 6 – 4 = 26 non-helper places. A suitable sentence is: The workshop is Friday in Room 5 with 26 places after the helper reservation, according to B and the capacity **calculation**.

The free bus is unsupported and must be removed or verified separately. The draft's 30 is total capacity before reservation.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible **trace**.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Name the operation and its evidence

For each task name the main operation and a suitable check: find the room listed in N2; work out six tables of four places minus three reserved places; suggest an inviting title for the event.

Can an application combine these operations?

Room: **retrieval**, checked against the right source N2 and event. Remaining places: **calculation**, checked as 6 × 4 − 3 = 21 with correct **input**s. Title: **generation**, checked for the requested style and unsupported factual details.

An application can combine them; the labels describe operations rather than exclusive types of application.

W2 Separate a correct sum from a correct answer

A calculator correctly returns 24 for 6 × 4. A pupil reports this as the number of non-helper places available. Explain the problem and write the correct expression, result and meaning.

24 is total capacity before the three helper places are reserved. The requested remaining count is 6 × 4 − 3 = 21 non-helper places.

The calculator can execute the supplied expression correctly while that expression omits a required step.

W3 Inspect a record and a draft

N1 lists Room 2. N2 explicitly replaces N1 for this event and lists Wednesday in Room 4. A draft says Thursday in Room 2 with free snacks. Identify the supported corrections and the unsupported detail.

Explain why finding N1 successfully would not settle the current-room question.

Use Wednesday and Room 4 from N2. Free snacks are unsupported and should be removed or verified with a suitable source.

N1 is an existing record but is superseded for this event, so retrieving it does not establish the current room.

W5 Do not infer a method from its output

Two applications both display “26 places remain.” Only one provides record B and a **trace** showing 5 × 6 – 4. What can and cannot be concluded from their matching sentences?

Describe how you would check the numerical claim without relying on either application’s assurance.

The visible numerical **output**s match, but that alone does not reveal the second application’s method or prove that either checked the right task. Inspect current B, confirm the **input**s and independently **calculate** 5 × 6 – 4 = 26.

The first **trace** supports inspection; a verbal assurance of checking is not equivalent to inspectable **evidence**.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

Generate the poem, **calculate** 25, **retrieve** the actual passage. Inventing a fluent passage is not **retrieval**.

Additional transfer practice question

Choose a method for each task: invent a dragon story; find the library's posted hours; compute 7 plus 5. Explain why the methods differ.

Reasoned practice answer

Generate the story, **retrieve** the posted hours, and **calculate** 12. Creativity, source lookup and arithmetic require different **evidence** and checking.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions

Criterion	0 points	1 point	2 points
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “Generating plausible text is the same as retrieving a fact.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Generate a motto and inspect suitability; **retrieve** and cite the actual handbook passage for the meeting point; **calculate** 84/7=12 stickers each and check 12×7=84. All need task-appropriate checks.

A fluent invented location cannot substitute for the handbook.

Explanations for the additional workbook examples

W1 Name the operation and its evidence

W2 Separate a correct sum from a correct answer

W3 Inspect a record and a draft

W4 Trace a new library workshop

Current fictional record B lists a library workshop on Friday in Room 5: five tables with six places each, with four places reserved for helpers. B replaces old record A, which lists Room 3.

A draft says “Thursday in Room 3, with 30 places remaining and a free bus home.” **Retrieve** the current day and room, **calculate** the non-helper places, write a corrected factual sentence and identify the detail that cannot be supported by B.

Retrieve Friday and Room 5 from B. **Calculate** 5 × 6 − 4 = 26 non-helper places. A suitable sentence is: The workshop is Friday in Room 5 with 26 places after the helper reservation, according to B and the capacity **calculation**.

The free bus is unsupported and must be removed or verified separately. The draft’s 30 is total capacity before reservation.

W5 Do not infer a method from its output

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Label every sorter component as **generate**, **retrieve**, **calculate**, fixed rule, learned **prediction**, or **human decision**.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. **Retrieve** this record next week rather than restarting the project.

Use the sorter’s approved catalogue for facts, explicit **calculation**s for scores and checked prose for explanations.

Support while preserving the reasoning goal

Use physical cards, spoken explanations and a three-stage drawing. Read aloud without giving the answer. Stretch: compare two **mechanism**s with the same visible **output** and explain what **evidence** distinguishes them.

Use three coloured cards for recorded fact, derived quantity and suggested wording. Work with counters for total and reserved places.

Extension: explain which steps should be repeated if only the room changes versus if the number of tables changes.

W4 requires Friday, Room 5, record B, 5 × 6 − 4 = 26, a consistent corrected sentence and rejection of the unsupported free bus.

After support change B to reserve seven helper places: require 23 remaining with the same day and room, and continued rejection of the transport claim.

A numerical result without its source and expression is incomplete **evidence**.

Plan the delayed check

Suggested return points in this level: ai-1/26, ai-1/28, ai-1/31. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 26: Prompts Are Design Instructions. **Retrieve** today's **evidence** and identify what carries forward and what is new.

Week 26 turns a vague request into a clear prompt with supplied **evidence**, **output** requirements and tests.

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<summary>Earlier wording, alternatives and picture transcripts</summary>

Connect and extend learning

Helpful Classroom Sorter

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case
</details>

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Distinguish retrieving a record, calculating from **input**s and generating wording

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: A calculator correctly returns 24 for 6 × 4. A pupil reports this as the number of non-helper places available.

Explain the problem and write the correct expression, result and meaning.

Reasoned practice answer: 24 is total capacity before the three helper places are reserved. The requested remaining count is 6 × 4 − 3 = 21 non-helper places.

The calculator can execute the supplied expression correctly while that expression omits a required step.

Check every learner before discussion: Use the worked case for **Generation retrieval** and **calculation**. Proposed explanation: “Retrieving a record proves it is current”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Check event and version; N1 is superseded by N2 in this example.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Reserve four places instead of two, keeping the room and day unchanged. Require a **prediction** and a reason before comparing. Expected result: Eight learner places remain.

Reason: 3×4−4=8; this changes capacity for learners, not the **retrieve**d date or location.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, **trace**/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

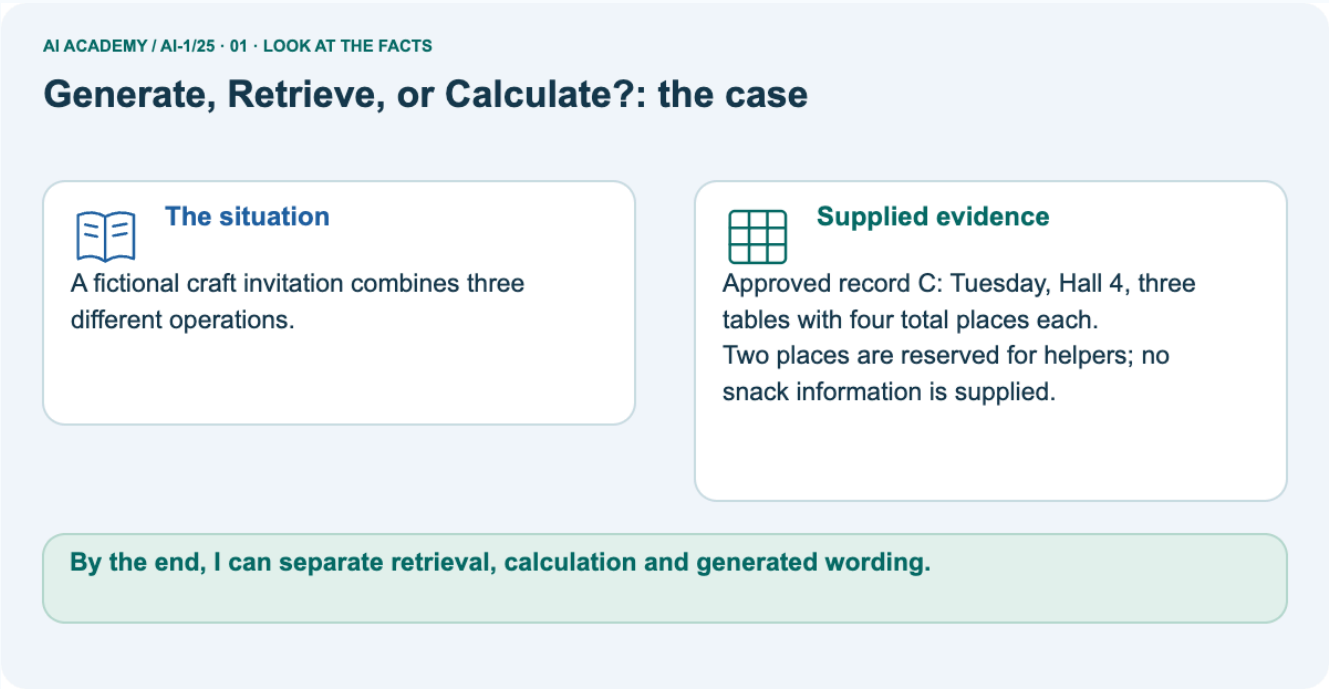
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Generate, Retrieve, or Calculate?: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

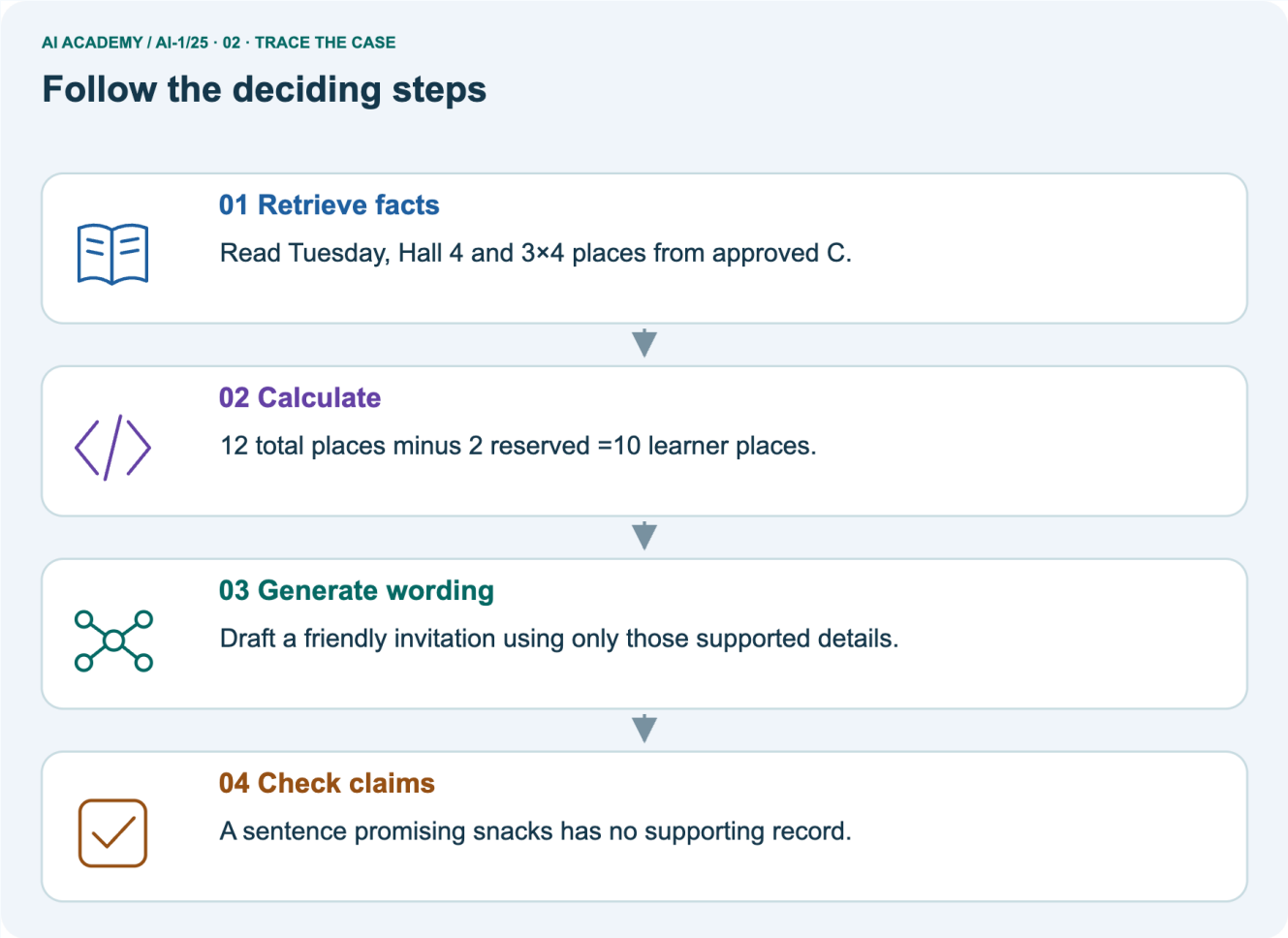
A fictional craft invitation combines three different operations. Supplied facts: Approved record C: Tuesday, Hall 4, three tables with four total places each.

Two places are reserved for helpers; no snack information is supplied.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



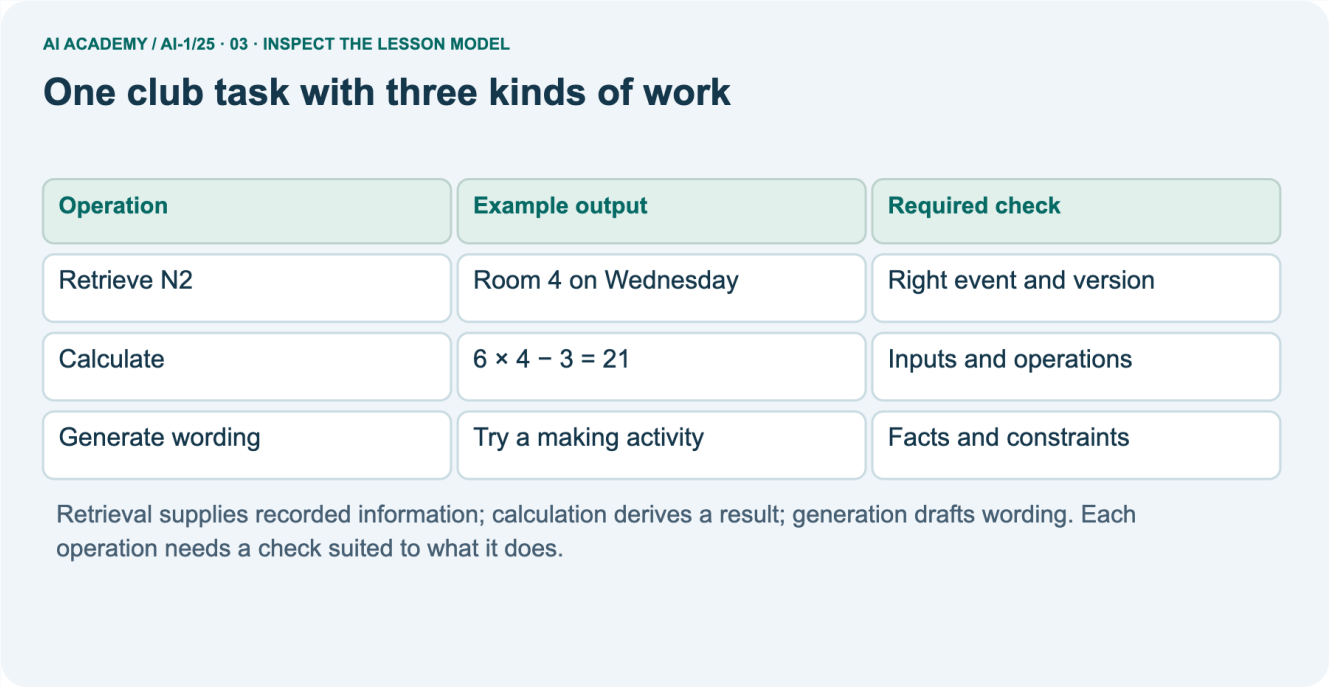
Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieve facts: Read Tuesday, Hall 4 and 3×4 places from approved C. → **Calculate**: 12 total places minus 2 reserved =10 learner places. → **Generate** wording: Draft a friendly invitation using only those supported details. → Check claims: A sentence promising snacks has no supporting record.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. One club task with three kinds of work



AI ACADEMY / AI-1/25 · 03 · INSPECT THE LESSON MODEL

One club task with three kinds of work

Operation	Example output	Required check
Retrieve N2	Room 4 on Wednesday	Right event and version
Calculate	6 × 4 − 3 = 21	Inputs and operations
Generate wording	Try a making activity	Facts and constraints

Retrieval supplies recorded information; calculation derives a result; generation drafts wording. Each operation needs a check suited to what it does.

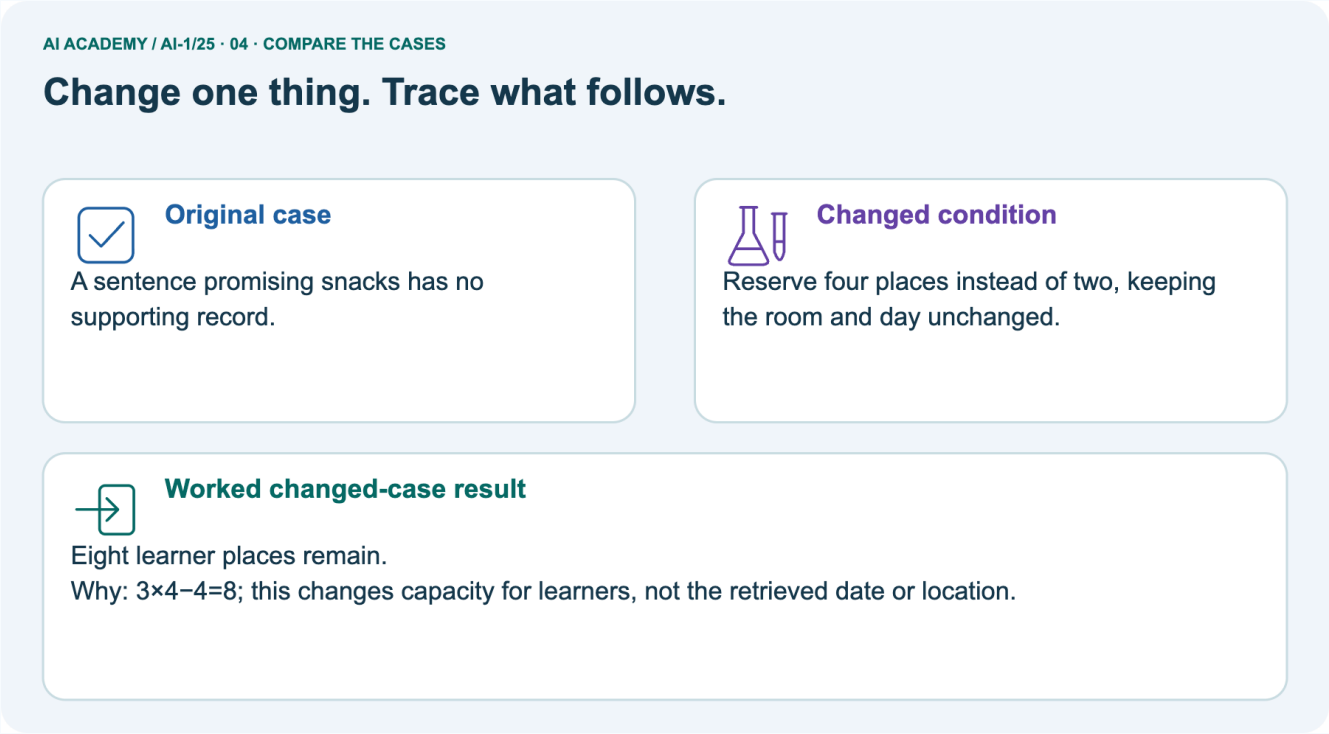
Follow the labelled picture with the matching explanation. Open the vector diagram

One club task with three kinds of work. **Retrieve** N2 | Room 4 on Wednesday | Right event and version. **Calculate** | 6 × 4 − 3 = 21 | **Input**s and operations. **Generate** wording | Try a making activity | Facts and constraints

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: A sentence promising snacks has no supporting record. Changed condition: Reserve four places instead of two, keeping the room and day unchanged. Expected result: Eight learner places remain.

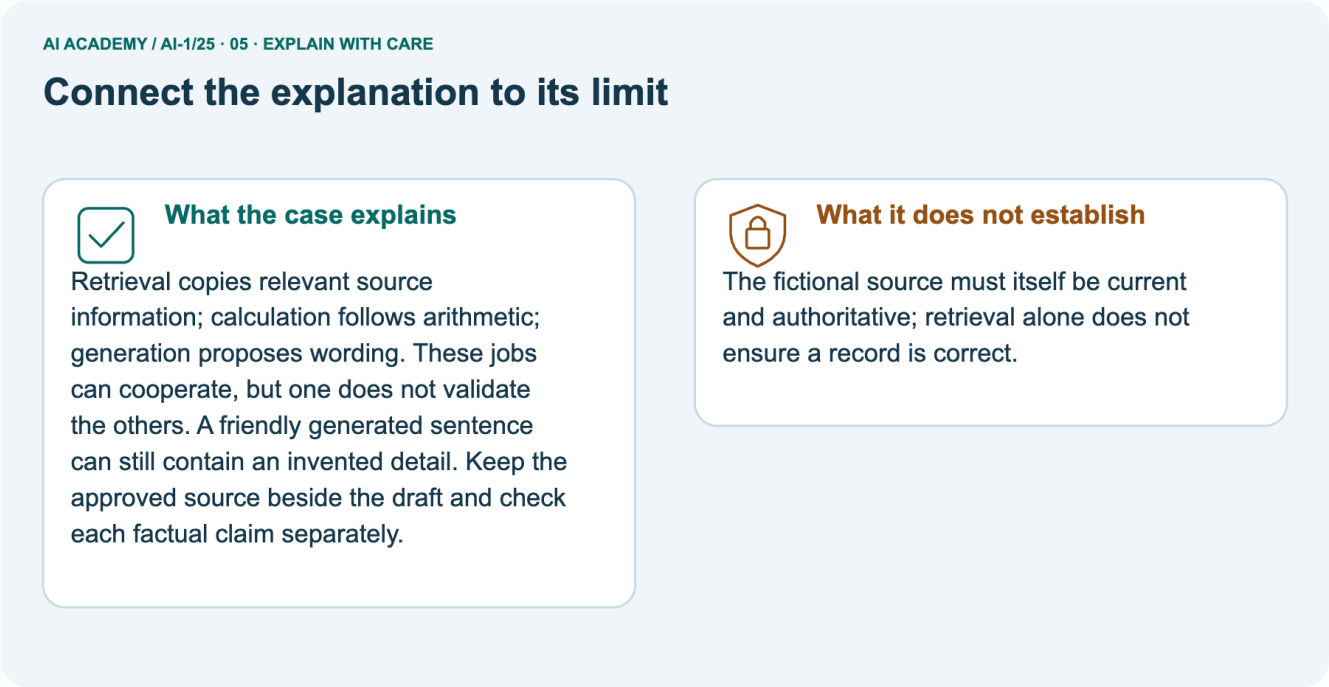
Reason: 3×4−4=8; this changes capacity for learners, not the **retrieve**d date or location.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: 3×4−4=8; this changes capacity for learners, not the **retrieve**d date or location.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Retrieval copies relevant source information; **calculation** follows arithmetic; **generation** proposes wording. These jobs can cooperate, but one does not validate the others.

A friendly **generate**d sentence can still contain an invented detail. Keep the approved source beside the draft and check each factual claim separately.

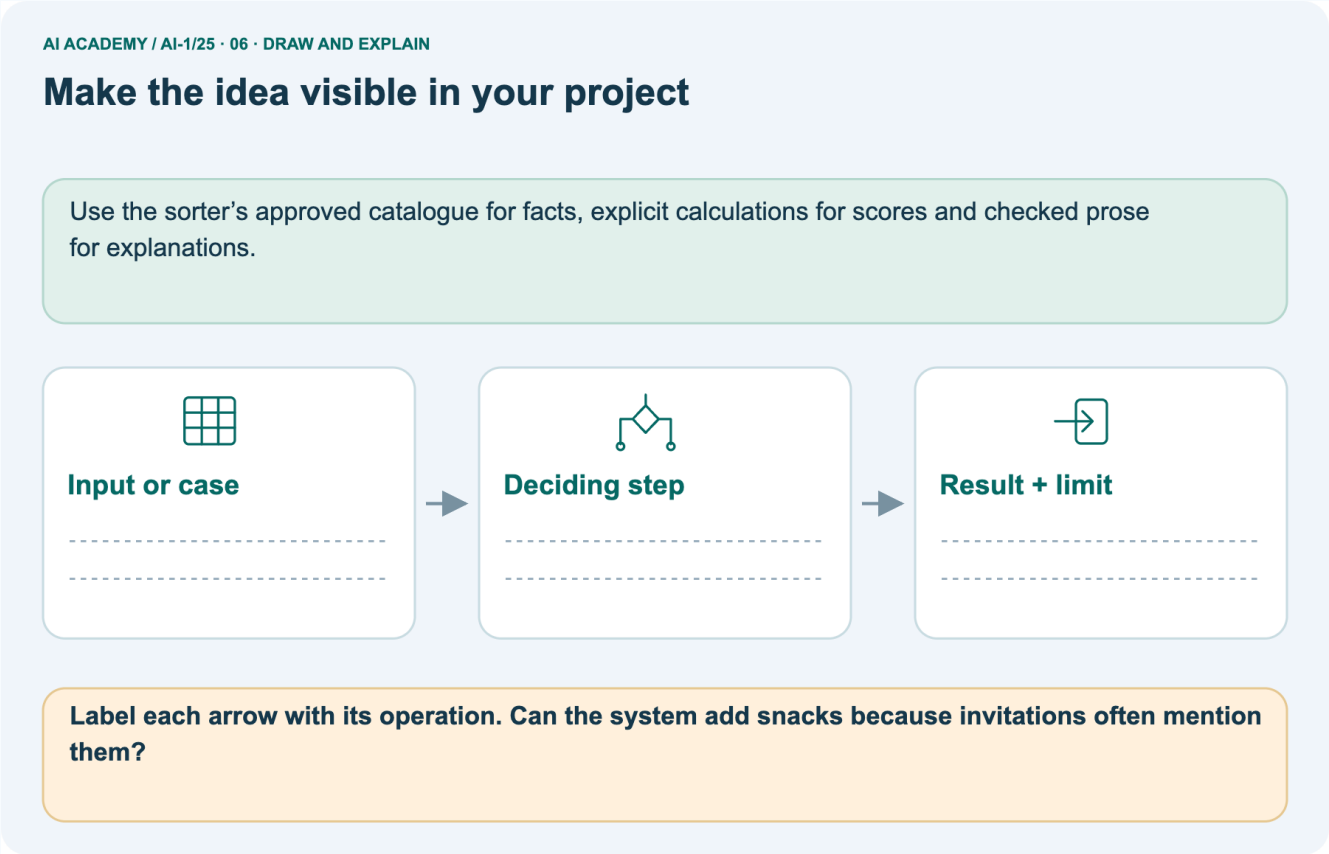
Limit: The fictional source must itself be current and authoritative; **retrieval** alone does not ensure a record is correct.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: The fictional source must itself be current and authoritative; **retrieval** alone does not ensure a record is correct.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use the sorter's approved catalogue for facts, explicit **calculation**s for scores and checked prose for explanations. Draw your **input**, deciding step, result and limit. Supported question: Label each arrow with its operation.

Can the system add snacks because invitations often mention them?

Draw your project **trace** and justify the deciding step using **evidence**.

Teacher prompt: Draw your project **trace** and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The arrows are **retrieval**, **calculation** and **generation**. No: snacks need confirmation; familiar wording is not **evidence**.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Generate, Retrieve, or Calculate?: the case

02. Follow the deciding steps

03. One club task with three kinds of work

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.